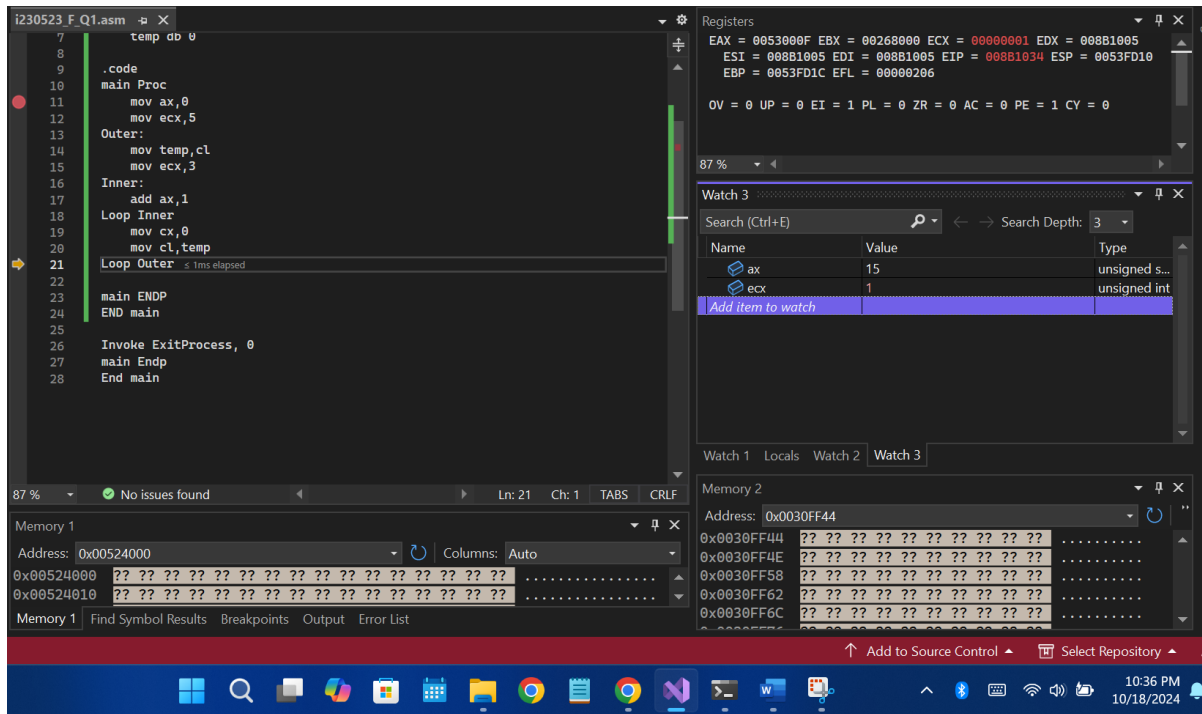


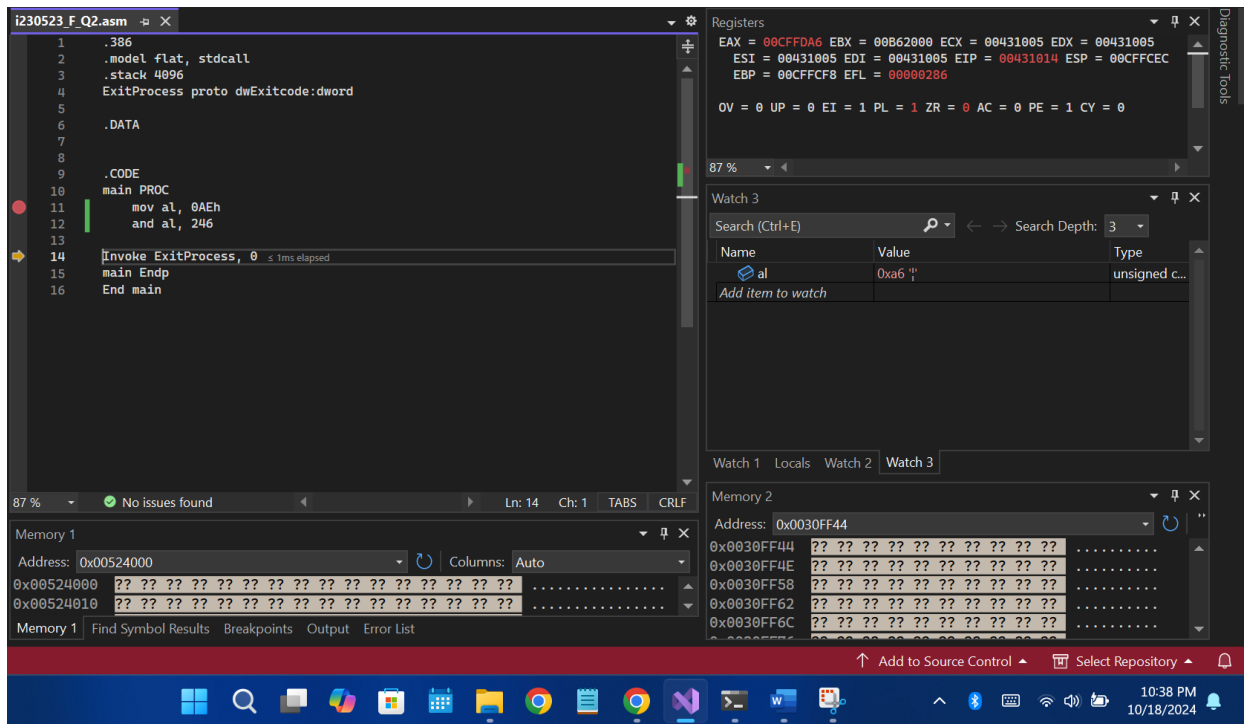
COAL Assignment-2

Code Screenshots

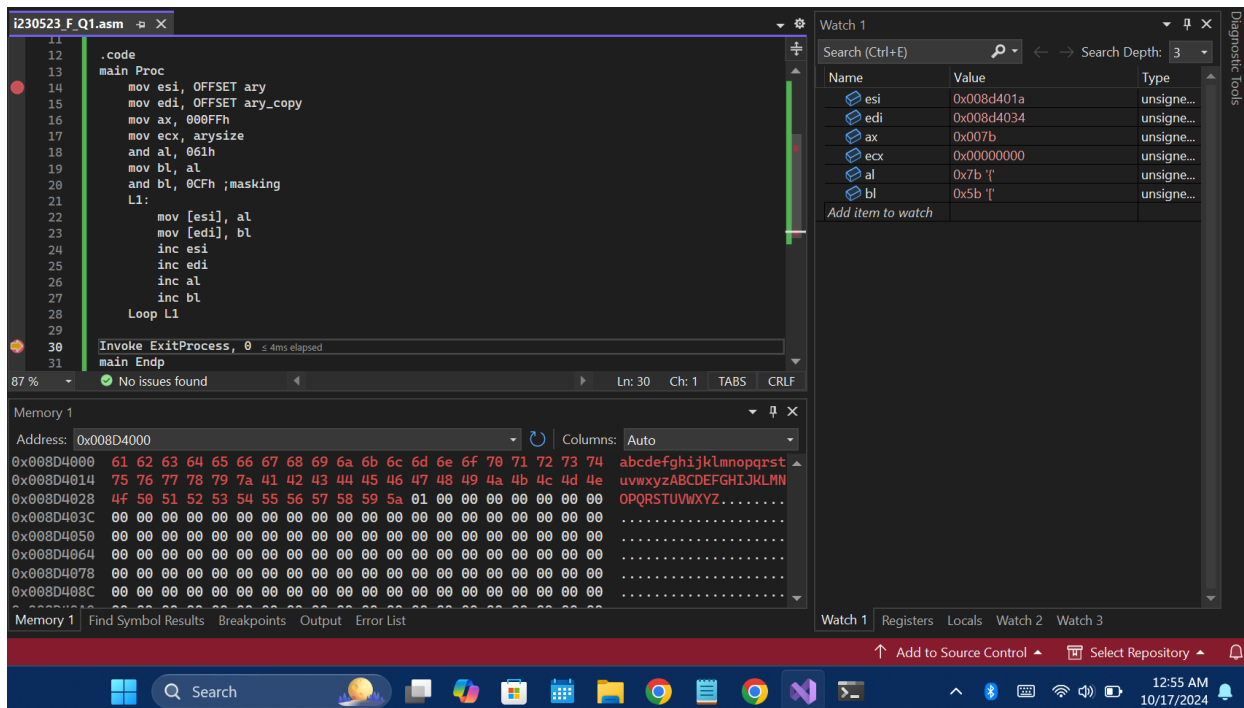
Question-1:



Question-2:



Question-3:



Question-4:

Assembly code for `i230523_F_Q4.asm`:

```
10  endmem db 1
11
12  .code
13  main Proc
14      mov esi, OFFSET ary
15      mov edi, OFFSET ary_copy
16      mov ax, 0041h
17      mov ecx, arysize
18      mov bl, al
19      or bl, 00100000b
20      L1:
21          mov [esi], al
22          mov [edi], bl
23          inc esi
24          inc edi
25          inc al
26          inc bl
27      Loop L1
28
29  Invoke ExitProcess, 0  < 3ms elapsed
30  main Endp
31  End main
```

Watch 1:

Name	Value	Type
esi	0x007b401a	unsigned int
edi	0x007b4034	unsigned int
ax	0x005b	unsigned s...
ecx	0x00000000	unsigned int
al	0x5b 'I'	unsigned c...
bl	0x7b 'I'	unsigned c...

Memory 1:

Address	0x007B4000	Columns	Auto
0x007B4000	41 42 43 44 45 46 47 48 49 4a 4b 4c 4d 4e 4f 50	ABCDEF	GHIJKLMNOP
0x007B4010	51 52 53 54 55 56 57 58 59 5a 61 62 63 64 65 66	QRSTU	VWXYZabcde
0x007B4020	67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73 74 75 76	ghijk	lmnopqrstuv
0x007B4030	77 78 79 7a 01 00 00 00 00 00 00 00 00 00 00 00	wxyz	
0x007B4040	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00		

Question-5:

Assembly code for `i230523_F_Q5.asm`:

```
1  .386
2  .model flat, stdcall
3  .stack 4096
4  ExitProcess proto dwExitcode:dword
5
6  .DATA
7
8
9  .CODE
10 main PROC
11     mov al, 11100011b
12     or al, 00000100
13
14     Invoke ExitProcess, 0  < 1ms elapsed
15 main Endp
16 End main
```

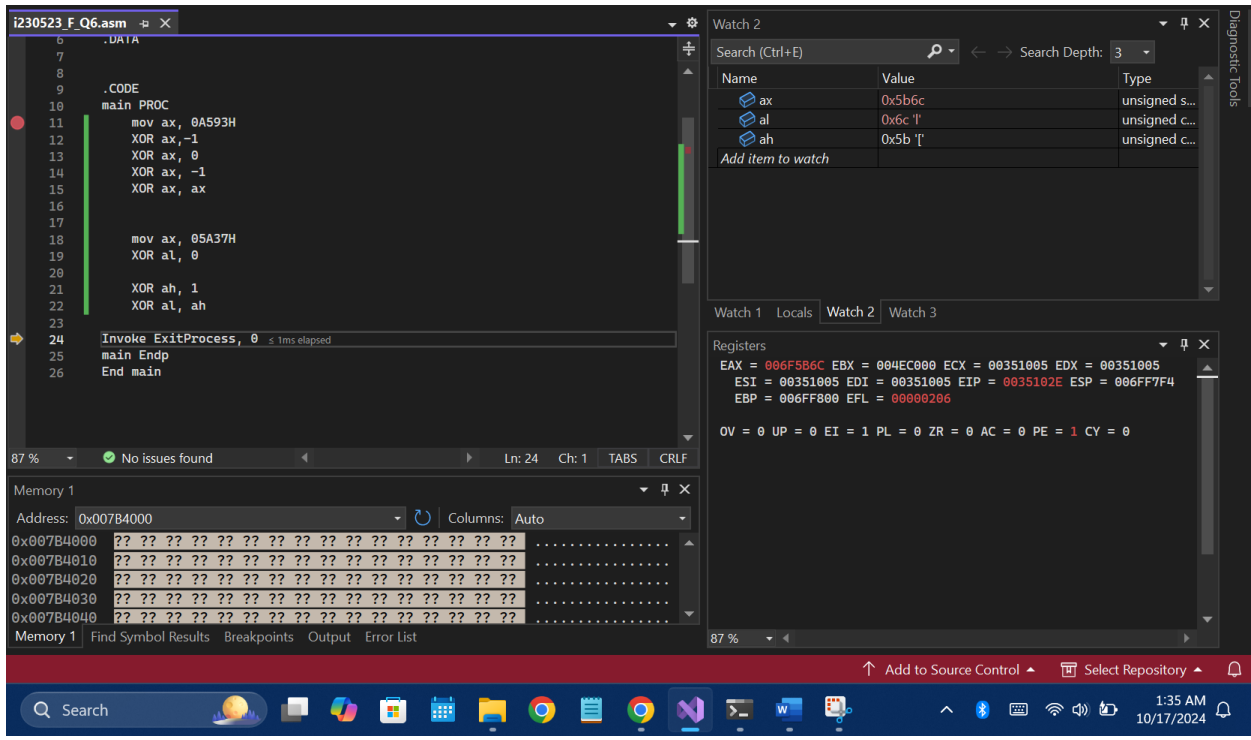
Registers:

Register	Value
EAX	008FFE7
EBX	006D2000
ECX	00051005
EDX	00051005
ESI	00051005
EDI	00051005
EIP	00051014
ESP	008FFF0C
EBP	008FFF18
EFL	00000286

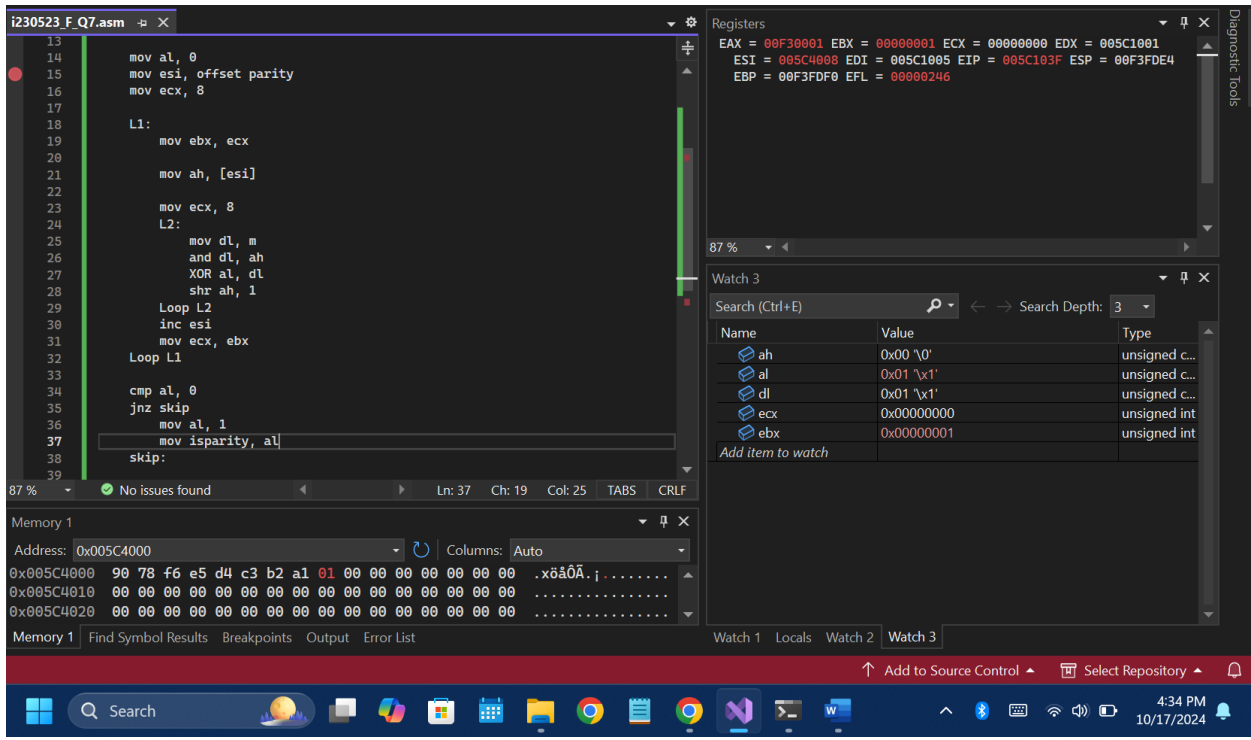
Memory 1:

Address	0x007B4000	Columns	Auto
0x007B4000	?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ??		
0x007B4010	?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ??		
0x007B4020	?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ??		
0x007B4030	?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ??		
0x007B4040	?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ??		

Question-6:



Question-7:



Question-8:

Assembly code for `i230523_F_Q8.asm` is shown. The code is at line 22, showing a loop with a jump instruction. The registers window shows the following values:

- EAX = 012FFC6E
- EBX = 011A9000
- ECX = 004E1005
- EDX = 004E1005
- ESI = 004E1005
- EDI = 004E1005
- EIP = 004E1022
- ESP = 012FFA34
- EBP = 012FFA40
- EFL = 00000283

The watch window shows the value of `ax` as `0xc6`. The memory window shows the address `0x00944000`.

Question-9:

Assembly code for `i230523_F_Q8.asm` is shown. The code is at line 32, showing a loop with a jump instruction. The registers window shows the following values:

- EAX = 00EFAC2F
- EBX = 00DAE000
- ECX = 004E1005
- EDX = 004E1005
- ESI = 004E1005
- EDI = 004E1005
- EIP = 004E1034
- ESP = 00EFF7A4
- EBP = 00EFF7B0
- EFL = 00000A93

The watch window shows the value of `ax` as `0xc2f`. The memory window shows the address `&parity` and an error message "Unable to evaluate the expression."

Question-10:

Assembly code for i230523_F_Q8.asm:

```
22 mov ax,07B1Ah
23 sub ax,0CEE8h
24 jc L_1
25 L_1: jz L_2
26 L_2: jo L_3
27 L_3: js L_4
28 L_4: jnp L_5
29 L_5:
30
31 ; ----- Q - 10 ----- ;
32 mov cx,0FBCDh
33 add cx,0E4AAh
34 jnc L_1
35 L_1: jnz L_2
36 L_2: jo L_3
37 L_3: js L_4
38 L_4: jnp L_5 ;ERROR: L_5 not defined
39 L_5:
40
41 ; ----- Q - 11 ----- ;
42 mov bx,0FABDh
43 add bx,0684Ah
44 jc L_1
45 L_1: jz L_2
46 L_2: jno L_3
47 L_3: jns L_4
48 L_4: jnp L_5
49 L_5: mov ah,04ch
50
51 ; ----- Q - 12 ----- ;
52 mov bx,0A5A5h
53 not bx
54 cmp bx,05A5Ah
55 jz I1
56 I1: je I2
57 I2: jnz I3
58 I3: jne I4
59 I4:
60
61 ; ----- Q - 13 ----- ;
62 mov ecx,5
63 mov ax,1
64 I_1:
65 inc ax
```

Registers:

EAX = 00EFAC2F EBX = 00DAE000 ECX = 004EE077 EDX = 004E1005
ESI = 004E1005 EDI = 004E1005 EIP = 004E1047 ESP = 00EFF7A4
EBP = 00EFF7B0 EFL = 00000297

OV = 0 UP = 0 EI = 1 PL = 1 ZR = 0 AC = 1 PE = 1 CY = 1

Watch 2:

Name	Value	Type
ax	0xac2f	unsigned s...
cx	0xe077	unsigned s...

Memory 1:

Address: &parity Columns: Auto

Unable to evaluate the expression.

Question-11:

Assembly code for i230523_F_Q8.asm:

```
39 L_5:
40
41 ; ----- Q - 11 ----- ;
42 mov bx,0FABDh
43 add bx,0684Ah
44 jc L_1
45 L_1: jz L_2
46 L_2: jno L_3
47 L_3: jns L_4
48 L_4: jp L_5
49 L_5: mov ah,04ch
50
51 ; ----- Q - 12 ----- ;
52 mov bx,0A5A5h
53 not bx
54 cmp bx,05A5Ah
55 jz I1
56 I1: je I2
57 I2: jnz I3
58 I3: jne I4
59 I4:
60
61 ; ----- Q - 13 ----- ;
62 mov ecx,5
63 mov ax,1
64 I_1:
65 inc ax
```

Registers:

EAX = 00EF4C2F EBX = 00DA6307 ECX = 004EE077 EDX = 004E1005
ESI = 004E1005 EDI = 004E1005 EIP = 004E105C ESP = 00EFF7A4
EBP = 00EFF7B0 EFL = 00000213

OV = 0 UP = 0 EI = 1 PL = 0 ZR = 0 AC = 1 PE = 0 CY = 1

Watch 2:

Name	Value	Type
ax	0x4c2f	unsigned s...
cx	0xe077	unsigned s...
bx	0x6307	unsigned s...

Memory 1:

Address: &parity Columns: Auto

Unable to evaluate the expression.

Question-12:

Assembly code (i230523_F_Q8.asm):

```
49 L_5: mov ah,04ch
50
51 ; ----- Q - 12 ----- ;
52 mov bx,0A5A5h
53 not bx
54 cmp bx,0A5A5h
55 jz I1
56 I1: je I2
57 I2: jnz I3
58 I3: jne I4
59 I4:
60
61 ; ----- Q - 13 ----- ;
62 mov ecx,5
63 mov ax,1
64 I_1:
65 inc ax
66 dec ecx
67 jcxz end_loop
68 jmp I_1
69 end_loop:
70
71
72 Invoke ExitProcess, 0
73 main Endp
74 End main
```

Registers:

EAX = 00DD4C2F EBX = 00F25A5A ECX = 004EE077 EDX = 004E1005
ESI = 004E1005 EDI = 004E1005 EIP = 004E1070 ESP = 00DDFD8C
EBP = 00DDFD98 EFL = 00000246

OV = 0 UP = 0 EI = 1 PL = 0 ZR = 1 AC = 0 PE = 1 CY = 0

Watch 2:

Name	Value	Type
ax	0x4c2f	unsigned s...
cx	0xe077	unsigned s...
bx	0x5a5a	unsigned s...

Memory 1:

Address: &parity Columns: Auto
Unable to evaluate the expression.

Question-13:

Assembly code (i230523_F_Q8.asm):

```
49 L_5: mov ah,04ch
50
51 ; ----- Q - 12 ----- ;
52 mov bx,0A5A5h
53 not bx
54 cmp bx,0A5A5h
55 jz I1
56 I1: je I2
57 I2: jnz I3
58 I3: jne I4
59 I4:
60
61 ; ----- Q - 13 ----- ;
62 mov ecx,5
63 mov ax,1
64 I_1:
65 inc ax
66 dec ecx
67 jcxz end_loop
68 jmp I_1
69 end_loop:
70
71
72 Invoke ExitProcess, 0
73 main Endp
74 End main
```

Registers:

EAX = 00DD0006 EBX = 00F25A5A ECX = 00000000 EDX = 004E1005
ESI = 004E1005 EDI = 004E1005 EIP = 004E1081 ESP = 00DDFD8C
EBP = 00DDFD98 EFL = 00000246

OV = 0 UP = 0 EI = 1 PL = 0 ZR = 1 AC = 0 PE = 1 CY = 0

Watch 2:

Name	Value	Type
ax	0x0006	unsigned s...
ecx	0x00000000	unsigned int

Memory 1:

Address: &parity Columns: Auto
Unable to evaluate the expression.

Question-14:

Not needed

Question-15:

```
11 Start:
12 mov swap,0 ;Reset swap flag to no swap
13 mov ebx,0 ;initialize array index to zero
14 loop1:
15 mov ax,[ebx+data] ;load number to ax
16 cmp ax,[ebx+data+2] ;compare with next number
17 ;no swap if already in order
18 jbe noswap ;Q-15
19 jge noswap ;Q-16
20
21 mov dx,[ebx+data+2] ;load second element in dx
22 mov [ebx+data+2],ax ;store first number in second
23 mov [ebx+data],dx ;store second number in first
24 mov swap,1 ;flag that a swap has been done
25 noswap:
26 add bx,2 ;advance bx to next index
27 cmp bx,18 ;are we at last index
28 jne loop1 ;if not then compare next 2
29
30 cmp swap,1 ;check if swap has been done
31 je start ;if yes then make another pass
32
33 INVOKE ExitProcess,0 ;3ms elapsed
```

Registers

EAX = 00FFFFFF EBX = 00000012 ECX = 000D1005 EDX = 00000011
ESI = 000D1005 EDI = 000D1005 EIP = 000D105B ESP = 00EFFE7C
EBP = 00EFFE88 EFL = 00000297

OV = 0 UP = 0 EI = 1 PL = 1 ZR = 0 AC = 1 PE = 1 CY = 1

Watch 2

Search (Ctrl+E) Search Depth: 3

Name	Value	Type
Add item to watch		

Memory 1

Address: 0x000D4000 Columns: Auto

0x000D4000	00 00 02 00 03 00 05 00 11 00 40 00 45 00 66 00	@.E.f.
0x000D4010	db ff ff 00 00 00 00 00 00 00 00 00 00 00 00	dyuy.....
0x000D4020	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0x000D4030	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0x000D4040	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0x000D4050	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	

Memory 1 Find Symbol Results Breakpoints Output Error List

Watch 1 Locals Watch 2 Watch 3

Add to Source Control Select Repository

Search

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Question-16:

```
10 main PROC
11 Start:
12 mov swap,0 ;Reset swap flag to no swap
13 mov ebx,0 ;initialize array index to zero
14 loop1:
15 mov ax,[ebx+data] ;load number to ax
16 cmp ax,[ebx+data+2] ;compare with next number
17 ;no swap if already in order
18 jbe noswap ;Q-15
19 jge noswap ;Q-16
20
21 mov dx,[ebx+data+2] ;load second element in dx
22 mov [ebx+data+2],ax ;store first number in second
23 mov [ebx+data],dx ;store second number in first
24 mov swap,1 ;flag that a swap has been done
25 noswap:
26 add bx,2 ;advance bx to next index
27 cmp bx,18 ;are we at last index
28 jne loop1 ;if not then compare next 2
29
30 cmp swap,1 ;check if swap has been done
31 je start ;if yes then make another pass
32
33
```

Registers

EAX = 008FFFFFFF EBX = 00000012 ECX = 00E11005 EDX = 00E10011
ESI = 00E11005 EDI = 00E11005 EIP = 00E1105B ESP = 008FFF60
EBP = 008FFF6C EFL = 00000297

OV = 0 UP = 0 EI = 1 PL = 1 ZR = 0 AC = 1 PE = 1 CY = 1

Watch 2

Search (Ctrl+E) Search Depth: 3

Name	Value	Type
Add item to watch		

Memory 1

Address: 0x00E14000 Columns: Auto

0x00E14000	66 00 45 00 40 00 11 00 05 00 03 00 02 00 00 00	f.E.@.....
0x00E14010	ff ff db ff 00 00 00 00 00 00 00 00 00 00 00 00	dyuy.....
0x00E14020	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0x00E14030	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0x00E14040	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	
0x00E14050	00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00	

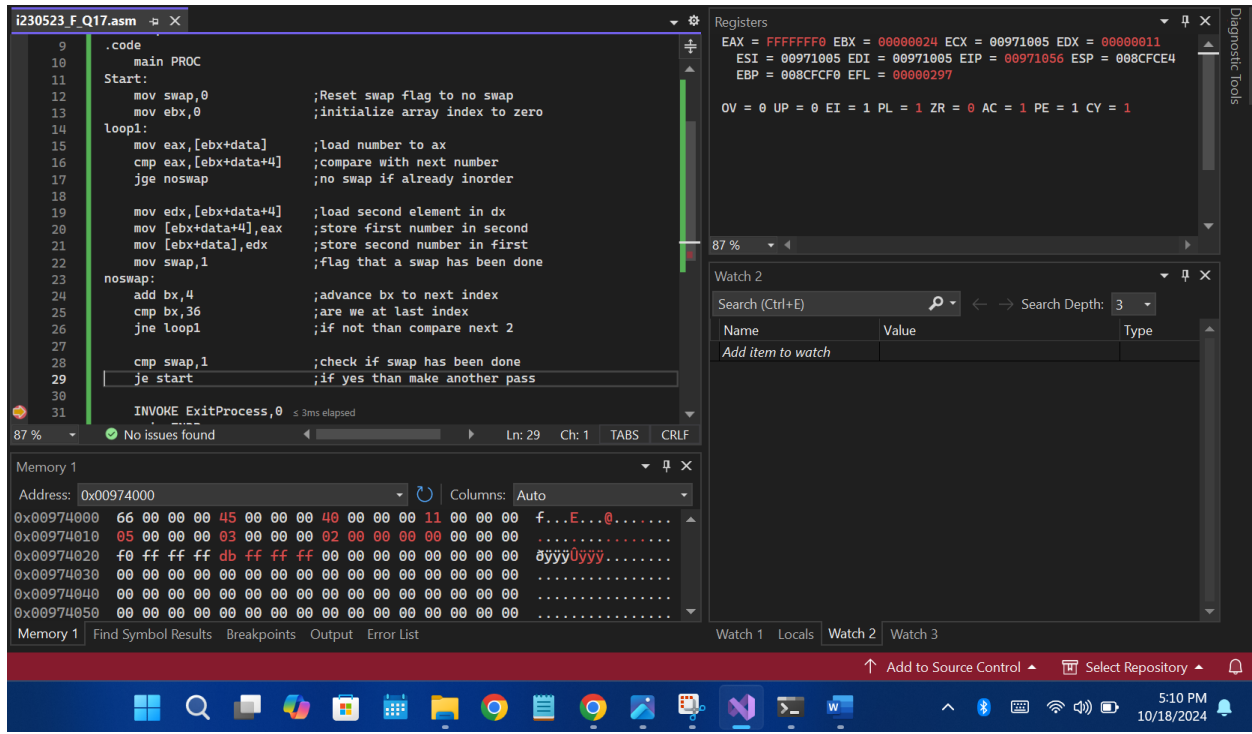
Memory 1 Find Symbol Results Breakpoints Output Error List

Watch 1 Locals Watch 2 Watch 3

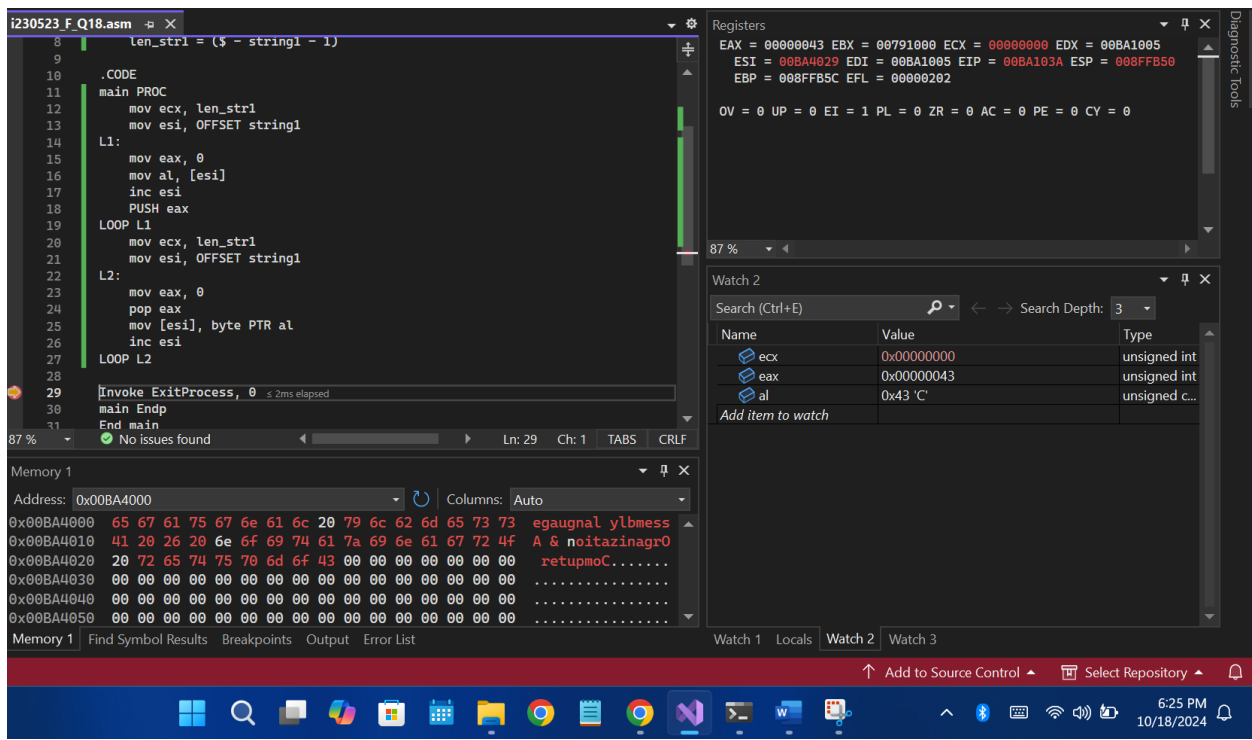
Add to Source Control Select Repository

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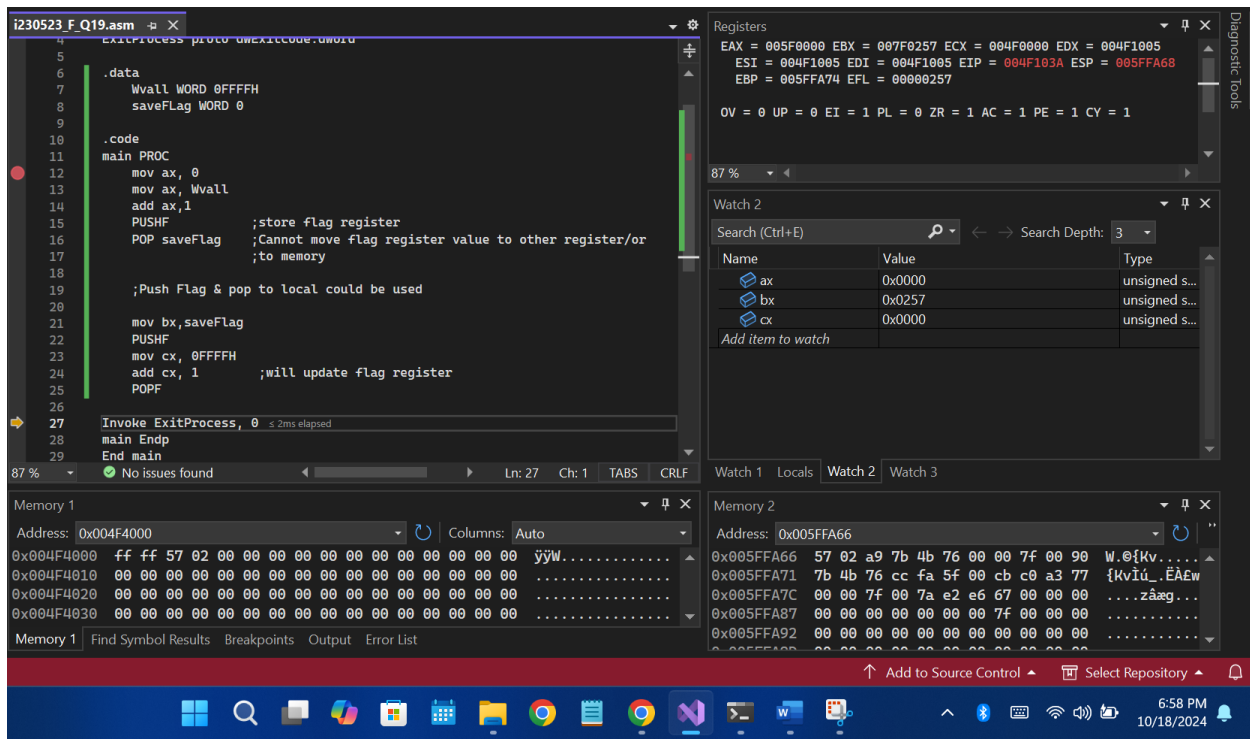
Question-17:



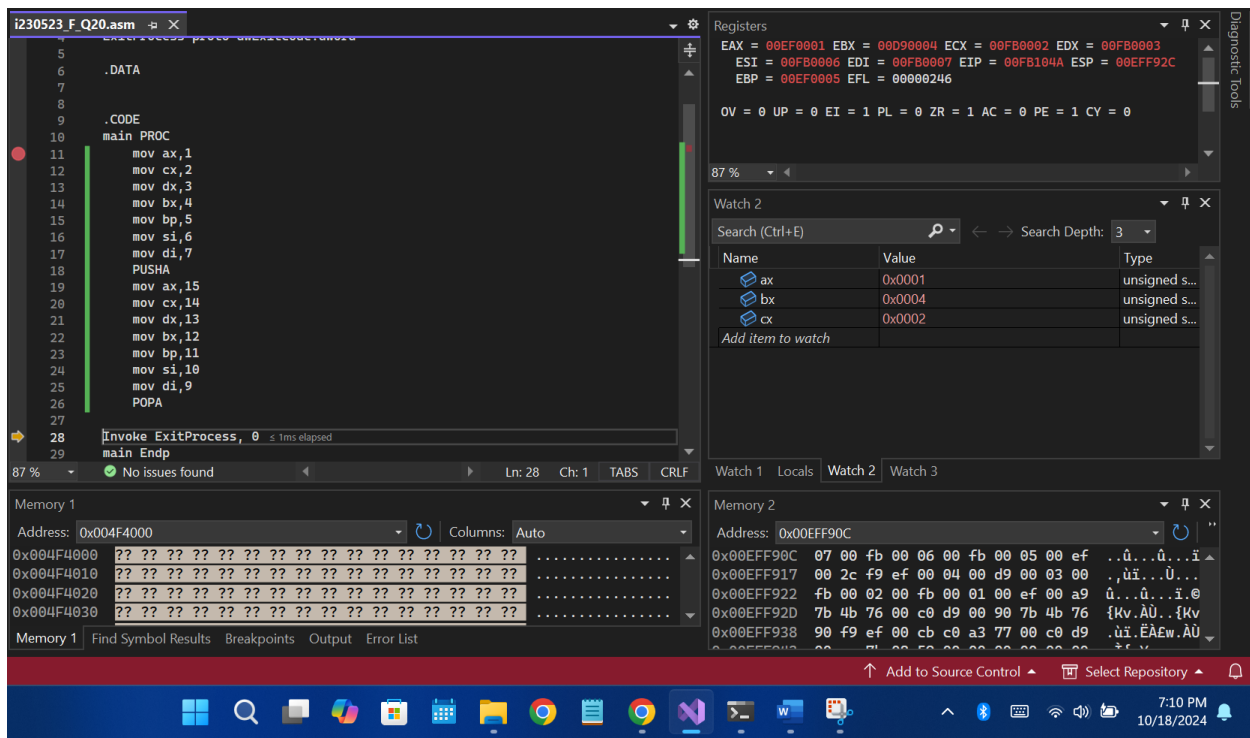
Question-18:



Question-19:



Question-20:



Question-21:

Assembly code for Question 21:

```
7  
8  
9 .CODE  
10 Sum PROC  
11     add ax,bx  
12     add ax,cx  
13     RET  
14 Sum ENDP  
15 Main PROC  
16     mov ax,1  
17     mov bx,2  
18     mov cx,3  
19     call sum  
20  
21     Invoke ExitProcess, 0  
22 main Endp  
23 End main
```

Registers:

EAX = 003B0006 EBX = 00450002 ECX = 00D00003 EDX = 00D0100A
ESI = 00D0100A EDI = 00D0100A EIP = 00D01034 ESP = 003BFBB4
EBP = 003BFB00 EFL = 00000206

OV = 0 UP = 0 EI = 1 PL = 0 ZR = 0 AC = 0 PE = 1 CY = 0

Watch 2:

Name	Value	Type
ax	0x0006	unsigned s...
bx	0x0002	unsigned s...
cx	0x0003	unsigned s...

Memory 1:

Address: 0x004F4000

Address	Value
0x004F4000	?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ??
0x004F4010	?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ??
0x004F4020	?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ??
0x004F4030	?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ??

Question-22:

Assembly code for Question 22:

```
4 ExitProcess proto dwExitcode:dword  
5  
6 .data  
7     val1 Dword 5  
8     val2 Dword 6  
9  
10 .code  
11 Sum PROC  
12     PUSH ebp  
13     mov ebp,esp  
14     mov ax,0  
15     add ax,[ebp+4]  
16     add ax,[ebp+6]  
17     POP ebp  
18     RET 4  
19 Sum ENDP  
20  
21 Main PROC  
22     Push val1  
23     Push val2  
24     call sum  
25  
26     Invoke ExitProcess, 0  
27 main Endp  
28 End main
```

Registers:

EAX = 010F10D7 EBX = 00F0C000 ECX = 0095100A EDX = 0095100A
ESI = 0095100A EDI = 0095100A EIP = 00951042 ESP = 010FF954
EBP = 010FF960 EFL = 00000206

OV = 0 UP = 0 EI = 1 PL = 0 ZR = 0 AC = 0 PE = 1 CY = 0

Watch 2:

Name	Value	Type
ax	0x10d7	unsigned s...
bx	0xc000	unsigned s...
cx	0x100a	unsigned s...

Memory 1:

Address: 0x004F4000

Address	Value
0x004F4000	?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ??
0x004F4010	?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ??
0x004F4020	?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ??
0x004F4030	?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ?? ??

Question-23:

N/A

Question-24:

The screenshot displays a debugger interface with the following components:

- Assembly View (Left):** Shows assembly code for a function named `Sortit`. The code includes instructions for comparing, moving, and swapping elements in an array. The current instruction is `je start` at line 33.
- Registers (Top Right):** Displays the current state of the CPU registers. For example, `EAX = 00000005`, `EBX = 00000004`, `ECX = 00200000`, `EDX = 00200011`, `ESI = 00204000`, `EDI = 0020100A`, `EIP = 00201078`, `ESP = 0030FF54`, `EBP = 0030FF44`, and `EFL = 00000212`.
- Watch (Middle Right):** A table showing the values of variables being watched. The table has columns for Name, Value, and Type.
- Memory (Bottom Right):** Displays the contents of memory at address `0x0030FF44`. The memory is shown in hexadecimal and ASCII format.

Name	Value	Type
eax	0x00000005	unsigned int
ebx	0x00000004	unsigned int
ecx	0x00200000	unsigned int
edx	0x00200011	unsigned int

Address	Value	Type
0x0030FF44	60 ff 30 00 00 00 00 04 00	'y0.....
0x0030FF4E	00 00 00 40 20 00 a9 7b 4b 76	...0 .0{Kv
0x0030FF58	00 20 49 00 90 7b 4b 76 b8 ff	. I..{Kv,y
0x0030FF62	30 00 cb c0 a3 77 00 20 49 00	0.EAew. I.
0x0030FF6C	82 8a 6c 93 00 00 00 00 00 00	.S1".....